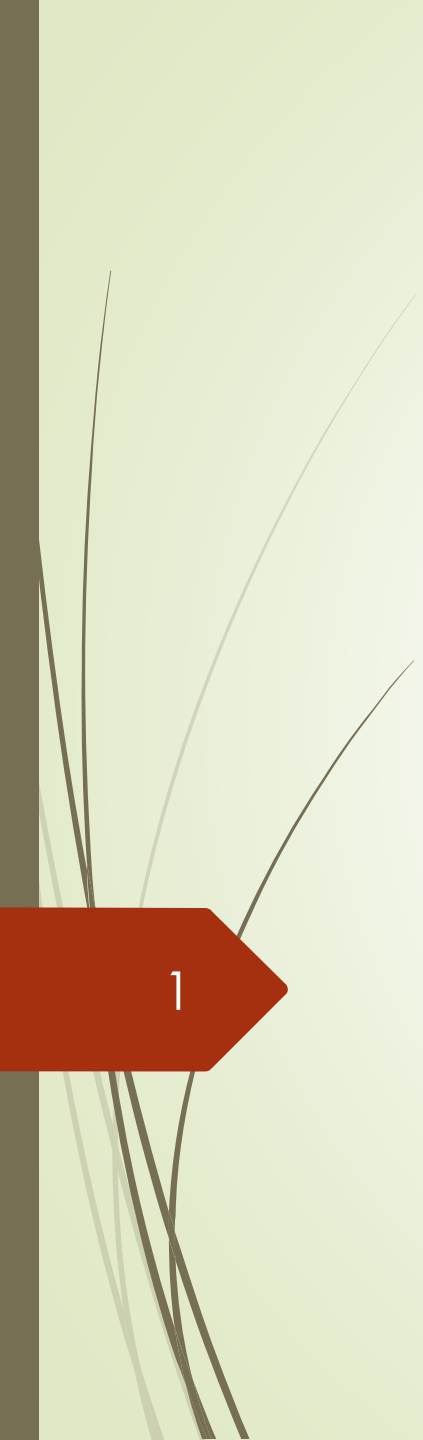




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Machine Learning

Lecture 7

Supervised learning Techniques and Algorithms –Bayesian Method

Lesson Outline

- Background and Introduction
- Bayes Theorem
- Examples using Bayes Theorem
- Naïve Bayes Classifiers

Bayes Theorem

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$$\text{Posterior} \rightarrow P(A | B) = \frac{P(A \cap B)}{P(B)} = \frac{\overset{\text{Likelihood}}{P(B | A)} \overset{\text{Prior}}{P(A)}}{\underset{\text{Marginal Likelihood}}{P(B)}}$$

➤ Task: determine the probability that incoming email was spam

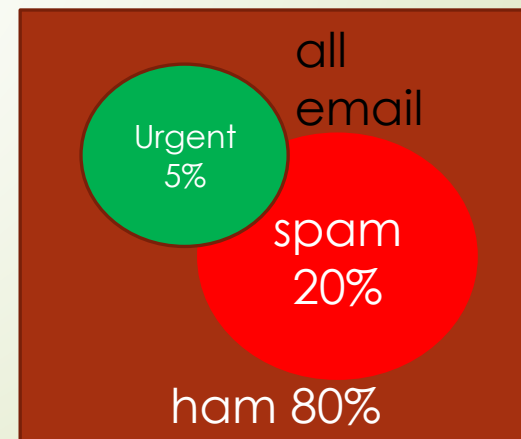
➤ First guess – prior probability – $P(\text{spam})$ or $P(A)$

➤ Additional information

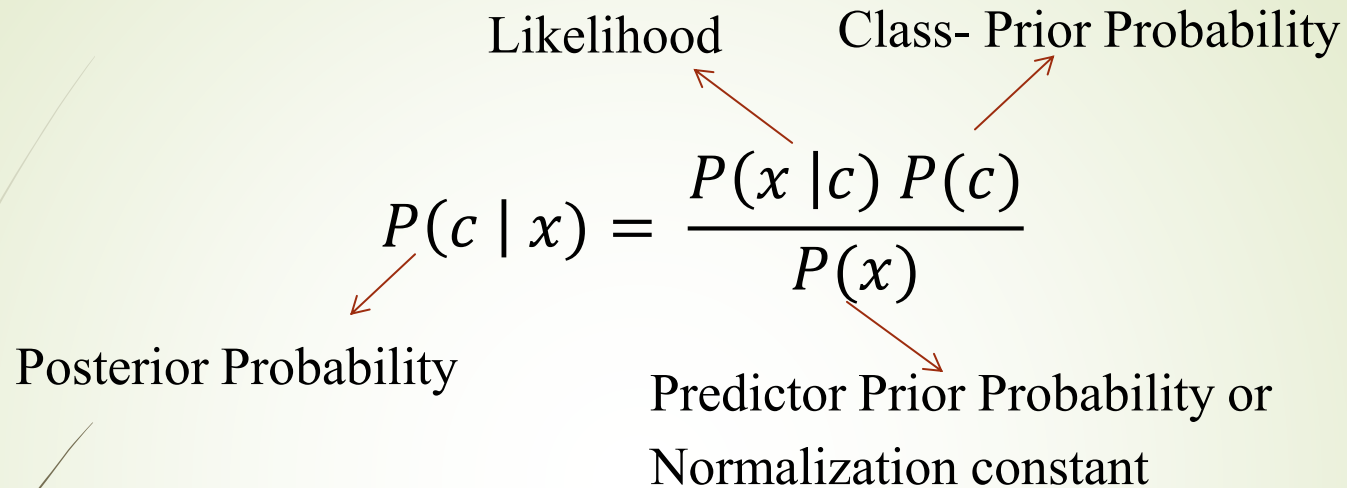
➤ A spam message contains the word “Urgent” – likelihood given by $P(\text{Urgent} | \text{spam})$ or $P(B | A)$

➤ Any message contains the word “Urgent” – marginal likelihood $P(\text{Urgent})$ or $P(B)$

➤ Now compute the posterior probability $P(\text{spam} | \text{Urgent})$ or $P(A | B)$



Bayes Theorem



The diagram shows the Bayes Theorem equation with four labels and arrows pointing to its components:

- Likelihood** points to $P(x | c)$
- Class- Prior Probability** points to $P(c)$
- Posterior Probability** points to $P(c | x)$
- Predictor Prior Probability or Normalization constant** points to $P(x)$

$$P(c | x) = \frac{P(x | c) P(c)}{P(x)}$$

Prior Probability: Describes the degree to which we believe the model accurately describes reality based on all of our prior information

Likelihood: Describes how well the model predicts the data

Normalization Constant: Makes the posterior density integrate to one

Posterior Probability: Shows the degree to which we believe a given model accurately describes the situation given the available data and all the prior information

Example Using Bayes Theorem

Day	outlook	Temperature	Humidity	Windy	Play
1	Sunny	Hot	High	FALSE	No
2	Sunny	Hot	High	TRUE	No
3	Overcast	Hot	High	FALSE	Yes
4	Rainy	Mild	High	FALSE	Yes
5	Rainy	Cool	Normal	FALSE	Yes
6	Rainy	Cool	Normal	TRUE	No
7	Overcast	Cool	Normal	TRUE	Yes
8	Sunny	Mild	High	FALSE	No
9	Sunny	Cool	Normal	FALSE	Yes
10	Rainy	Mild	Normal	FALSE	Yes
11	Sunny	Mild	Normal	TRUE	Yes
12	Overcast	Mild	High	TRUE	Yes
13	Overcast	Hot	Normal	FALSE	Yes
14	Rainy	Mild	High	TRUE	No



Example Using Bayes Theorem

Outlook			Temp			Humidity			Windy			Play		
	Yes	No		Yes	No		Yes	No		Yes	No		Yes	No
Sunny	2	3	Hot	2	2	High	3	4	False	6	2		9	5
Overcast	4	0	Mild	4	2	Normal	6	1	True	3	3			
Rainy	3	2	Cool	3	1									
Sunny	2/9	3/5	Hot	2/9	2/5	High	3/9	4/5	False	6/9	2/5		9/14	5/14
Overcast	4/9	0/5	Mild	4/9	2/5	Normal	6/9	1/5	True	3/9	3/5			
Rainy	3/9	2/5	Cool	3/9	1/5									

Example: $P(\text{Temperature}=\text{cool} \mid \text{Play}=\text{Yes})=3/9$

Example: $P(\text{Outlook}=\text{Rainy} \mid \text{Play}=\text{No})=2/5$

What is the probability of playing knowing in this condition?

Outlook	Temperature	Humidity	Windy	Play
Sunny	Cool	High	True	?

Let's call this condition X and the Playing, C

So Play or not??!!



Example

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$$P(\text{Play} = \text{Yes} | X) = \frac{P(X | \text{Play} = \text{Yes})P(\text{Play} = \text{Yes})}{P(X)}$$

$$P(X | \text{Play} = \text{Yes})P(\text{Play} = \text{Yes}) = \frac{2}{9} * \frac{3}{9} * \frac{3}{9} * \frac{3}{9} * \frac{9}{14} = 0.0053$$

$$P(X | \text{Play} = \text{No})P(\text{Play} = \text{No}) = \frac{3}{5} * \frac{1}{5} * \frac{4}{5} * \frac{3}{5} * \frac{5}{14} = 0.0206$$

If all the weather condition that we considered are independent from each other, then:

$$P(X) = P(\text{Outlook} = \text{sunny}) * P(\text{Temperature} = \text{Cool}) * P(\text{Humidity} = \text{High}) * P(\text{Windy} = \text{True})$$

$$P(X) = \frac{5}{14} * \frac{4}{14} * \frac{7}{14} * \frac{6}{14} = 0.02186$$

Remember:
$$P(c | x) = \frac{P(x | c) P(c)}{P(x)}$$

$$P(\text{Play} = \text{Yes} | X) = \frac{P(X | \text{Play} = \text{Yes})P(\text{Play} = \text{Yes})}{P(X)} = \frac{0.0053}{0.02186} = 0.2424$$

$$P(\text{Play} = \text{No} | X) = \frac{0.0206}{0.02186} = 0.9421$$

Naïve Bayes Classifiers

- A *priori* probability:
 - The probability of an event before evidence of that event is seen $P(H)$
- A *posteriori* probability:
 - The probability of an event after the evidence of that event is seen $P(H|E)$
- Assume independence- That's why we call it naïve!

$$P(E|H) = P(E_1|H) * P(E_2|H) * \dots * P(E_n|H)$$

And Bayes Theorem is:

$$\begin{aligned} P(H|E) &= P(E|H) * \frac{P(H)}{P(E)} \\ &= \frac{P(E_1|H)P(E_2|H) \dots P(E_n|H)P(H)}{P(E)} \end{aligned}$$

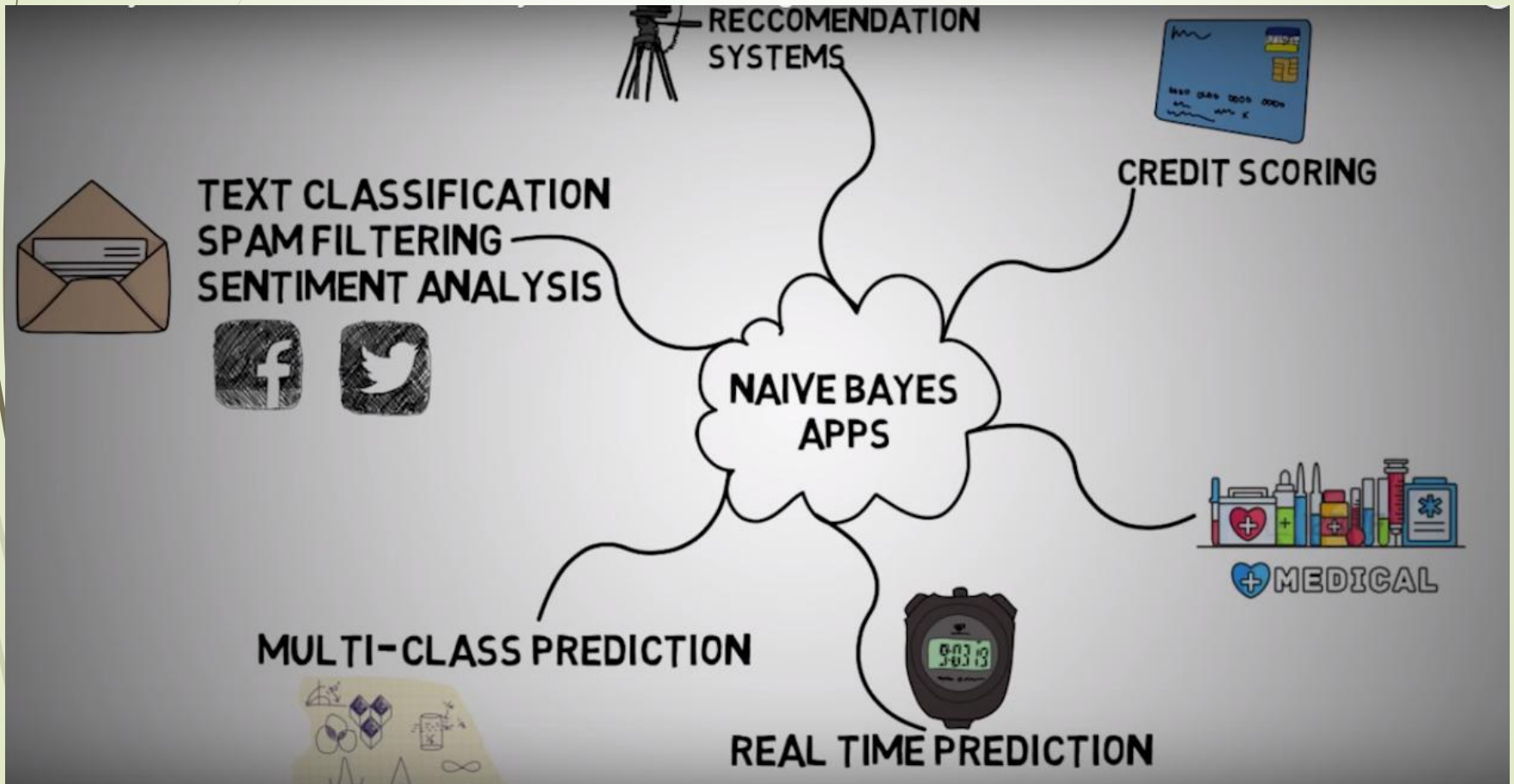
Was that essential to compute $P(X)$?

$$P(\text{Play} = \text{Yes} | X) = \frac{P(X | \text{Play} = \text{Yes})P(\text{Play} = \text{Yes})}{P(X)} = \frac{0.0053}{P(X)}$$

$$P(\text{Play} = \text{No} | X) = \frac{P(X | \text{Play} = \text{No})P(\text{Play} = \text{No})}{P(X)} = \frac{0.0206}{P(X)}$$

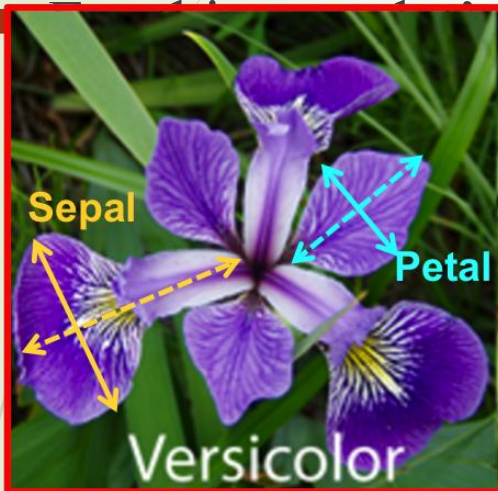
Remember that we don't actually have to calculate $P(X)$ because it is constant for all classes.

Naïve Bayes Classifier Applications



Multiple classes example

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	Petal_Length	Petal_Width	Sepal_Length	Sepal_Width	Species
0	1.4	0.2	5.1	3.5	0
1	1.4	0.2	4.9	3.0	0
2	1.3	0.2	4.7	3.2	0
3	1.5	0.2	4.6	3.1	0
4	1.4	0.2	5.0	3.6	0
5	1.7	0.4	5.4	3.9	0
6	1.4	0.3	4.6	3.4	0
7	1.5	0.2	5.0	3.4	0

	Petal_Length	Petal_Width	Sepal_Length	Sepal_Width	Species
count	150.000000	150.000000	150.000000	150.000000	150.000000
mean	3.758667	1.198667	5.843333	3.054000	1.000000
std	1.764420	0.763161	0.828066	0.433594	0.819232
min	1.000000	0.100000	4.300000	2.000000	0.000000
25%	1.600000	0.300000	5.100000	2.800000	0.000000
50%	4.350000	1.300000	5.800000	3.000000	1.000000
75%	5.100000	1.800000	6.400000	3.300000	2.000000
max	6.900000	2.500000	7.900000	4.400000	2.000000

Multiple classes example

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► Let's see what is going in our problem

Setosa | length and width information

$$P(c_1 | X) = \frac{P(X | c_1) P(c_1)}{P(X)}$$

petal length | Setosa

petal width | Setosa

sepal length | Setosa

sepal width | Setosa

$$P(c_1 | X) = \frac{P(x_1 | c_1) P(x_2 | c_1) P(x_3 | c_1) P(x_4 | c_1) P(c_1)}{P(X)}$$

Virginica | length and width information

$$P(c_2 | X) = \frac{P(x_1 | c_2) P(x_2 | c_2) P(x_3 | c_2) P(x_4 | c_2) P(c_2)}{P(X)}$$

Versicolor | length and width information

$$P(c_3 | X) = \frac{P(x_1 | c_3) P(x_2 | c_3) P(x_3 | c_3) P(x_4 | c_3) P(c_3)}{P(X)}$$

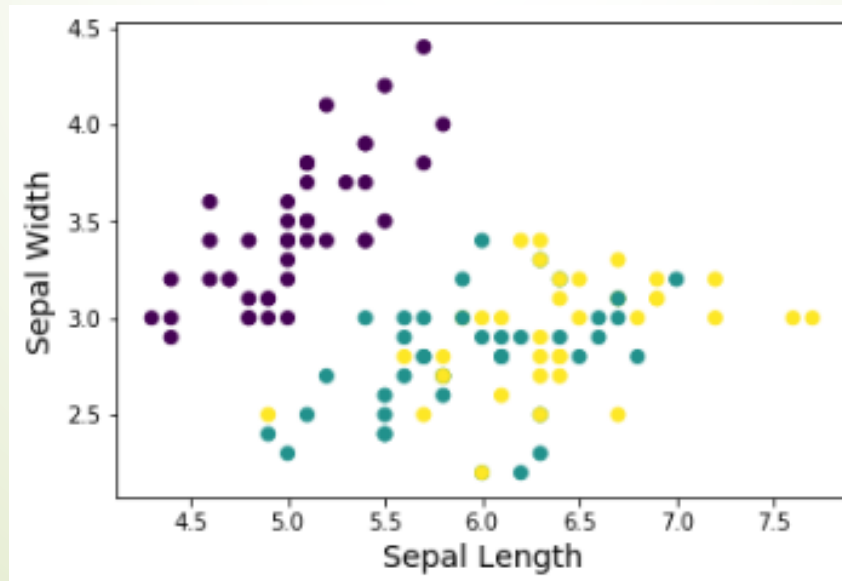
Split data to train and test

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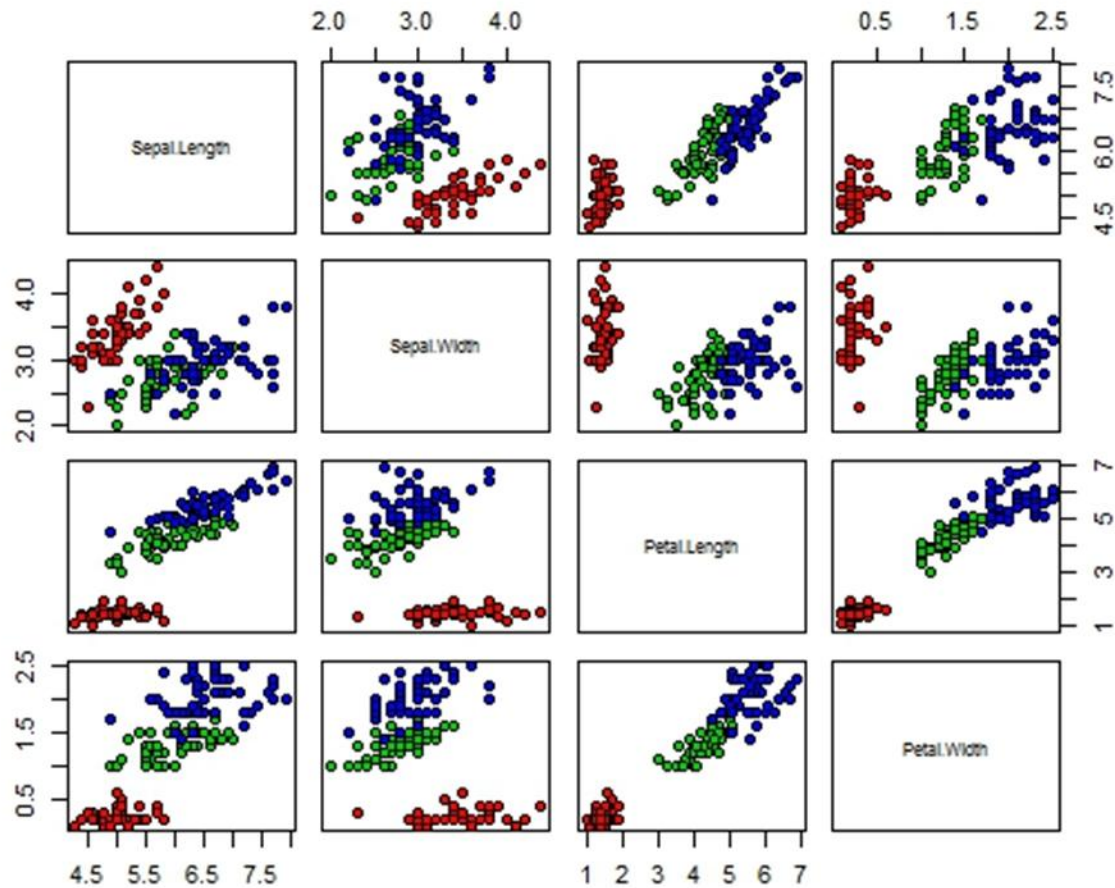
- ➡ Use same methodology as before to split data to train and test

```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y,
test_size=0.30, random_state=52)
```

- ➡ Now we have 70% of data for training and the rest for testing



Edgar Anderson's Iris Data



Train and evaluate the model

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- ➔ Now we can train and evaluate the model

```
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score
gnb = GaussianNB()
gnb = gnb.fit(X_train, y_train)
y_predict = gnb.predict(X_test)
accuracy_score(y_test, y_predict)*100
```

91.11111111111111

- ➔ One way of improving the accuracy is to change the prior probabilities

```
gnb = GaussianNB(priors=[.13, .74, .13])
```

97.77777777777777

- ➔ For discrete features we use *MultinomialNB()*

More information

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- We can get more information about the model

gnb.__dict__

```
{'class_count_': array([39., 33., 33.]),  
'class_prior_': array([0.37142857, 0.31428571, 0.31428571]),  
'classes_': array([0, 1, 2]),  
'priors': None,  
'sigma_': array([[0.1391453 , 0.13425378, 0.02390533, 0.00777121],  
                 [0.26049587, 0.06817264, 0.20022039, 0.02870524],  
                 [0.23825528, 0.08288338, 0.16128559, 0.08635446]]),  
'theta_': array([[5.03333333, 3.44358974, 1.43846154, 0.23076923],  
                 [5.93636364, 2.76969697, 4.20909091, 1.30909091],  
                 [6.34848485, 2.91212121, 5.35151515, 1.9969697 ]])}
```

- Here we can show likelihood of features
 - theta_ : mean of each feature per class
 - sigma_ : variance of each feature per class

R- Iris Dataset

```
> library(caTools)
> # Make 70% of data for training and the rest for testing
> set.seed(1)
> iris$spl <- sample.split(iris, SplitRatio = 0.7)
> train <- subset(iris, iris$spl==TRUE)
> test <- subset(iris, iris$spl==FALSE)
> # drop the newly added column
> train1 <- train[-6]
> test1 <- test[-6]
> library(naivebayes)
```

R- Naïve Bayes model

```
> naive <- naive_bayes(Species ~ ., data=train1)
> (mat<- table(predict(naive, test1[, -5]), test1[, 5]))
```

	setosa	versicolor	virginica
setosa	16	0	0
versicolor	0	16	1
virginica	0	1	16

```
> (Accuracy <- sum(diag(mat))/sum(mat)*100)
[1] 96
```

Pros and Cons

➤ Pros

- It is easy and fast for classification
- If the events are independent, Naïve Bayes usually acts better and we need less training data
- It works better on categorical variables compared to numerical ones

➤ Cons

- If there is a category which is not seen in training, the probability of zero will be assigned that affects the final result (Zero frequency)
- Assumption of independent redactors in real world does not hold